LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 11: Timers, Difficulty and Game Over

Module 4 | 45 minutes | Time and difficulty | Grades 6–8 · ages 11–14

Objective: Use the timer and counters to limit time, increase difficulty as the game goes on, and end the game cleanly.

Success criteria

- I can make a countdown or time limit.
- I can increase difficulty with a variable.
- I can end a game clearly.

Key vocabulary

- **countdown**: A variable that goes down to zero.
- **difficulty curve**: How a game gets harder over time.
- **game over**: The end state of a game.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Clap ramp (5 min): clap with the teacher, getting faster every 5 claps — when do people drop out?
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Clock and ramp: learners follow the steps.
14 min	**Perform**	Code Lab: countdown, Code Lab: speed it up
9 min	**Probe**	Shape the challenge, activity
4 min	**Wrap-up**	Exit ticket: Which variable did you use to make the game harder?

Unplugged warm-up

Clap ramp (5 min): clap with the teacher, getting faster every 5 claps — when do people drop out?

Teaching the idea

Time-based mechanics via counter variables or the timer reporter; difficulty curves via speed variables used in movement; clean termination via a final message and stop all.

Analogy: A treadmill programme: starts slow, speeds up, ends with your stats.

Live demo: Play the dodger with a fixed speed (boring) and a ramp (exciting).

Questions to ask

Ask	Listen for
Where is the say block — inside or after the loop?	After.
How does speed affect the rock?	It's used in the move block.
What makes a difficulty curve fair?	Gradual increases.

Differentiation

Support
Provide the countdown; learners add the ramp.

Stretch
Show a high score with a variable that keeps the best result.

Common misconceptions

Misconception	How to fix it
Changing a speed variable speeds things up by itself	The movement must use the variable.
stop all is the same as game over	Show feedback first, then stop.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which variable did you use to make the game harder?

Homework
Sketch the difficulty curve of a game you play.

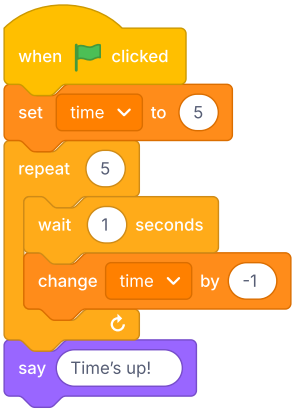
Answer key — Kitabash activities

1. Code Lab: countdown (perform · code lab)

Goals checked: Counts down to 0 · Starts at 5 · Announces the end

One solution:

Clock

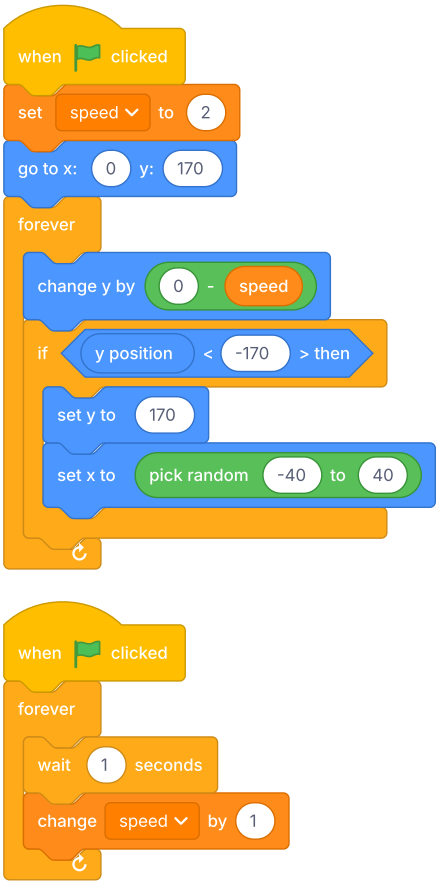


2. Code Lab: speed it up (perform · code lab)

Goals checked: Speed starts at 2 · Speed increases · Movement uses speed

One solution:

Rock



3. Shape the challenge (probe · ordering)

Order: Reset score, lives, speed and timer → Start slow so players learn the controls → Increase speed gradually → Lives or time run out → Show the final score and stop all

4. Explain your thinking (probe · free response)

Look for: speed, harder, end. **Example:** The rocks start falling slowly so players can learn the controls. Every 5 seconds the speed goes up by 1. The game ends after 60 seconds or when lives reach 0, and it shows the final score so players want to beat it.

Answer key — worksheet

1. **Key words:** countdown = A variable that goes down to zero. · difficulty curve = How a game gets harder over time. · game over = The end state of a game.
2. **Plan your code:** Goals: Counts down to 0; Starts at 5; Announces the end.
3. **Plan your code:** Goals: Speed starts at 2; Speed increases; Movement uses speed.

4. **Shape the challenge:** 1 Reset score, lives, speed and timer · 2 Start slow so players learn the controls · 3 Increase speed gradually · 4 Lives or time run out · 5 Show the final score and stop all
5. **Explain your thinking:** The rocks start falling slowly so players can learn the controls. Every 5 seconds the speed goes up by 1. The game ends after 60 seconds or when lives reach 0, and it shows the final score so players want to beat it.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.